

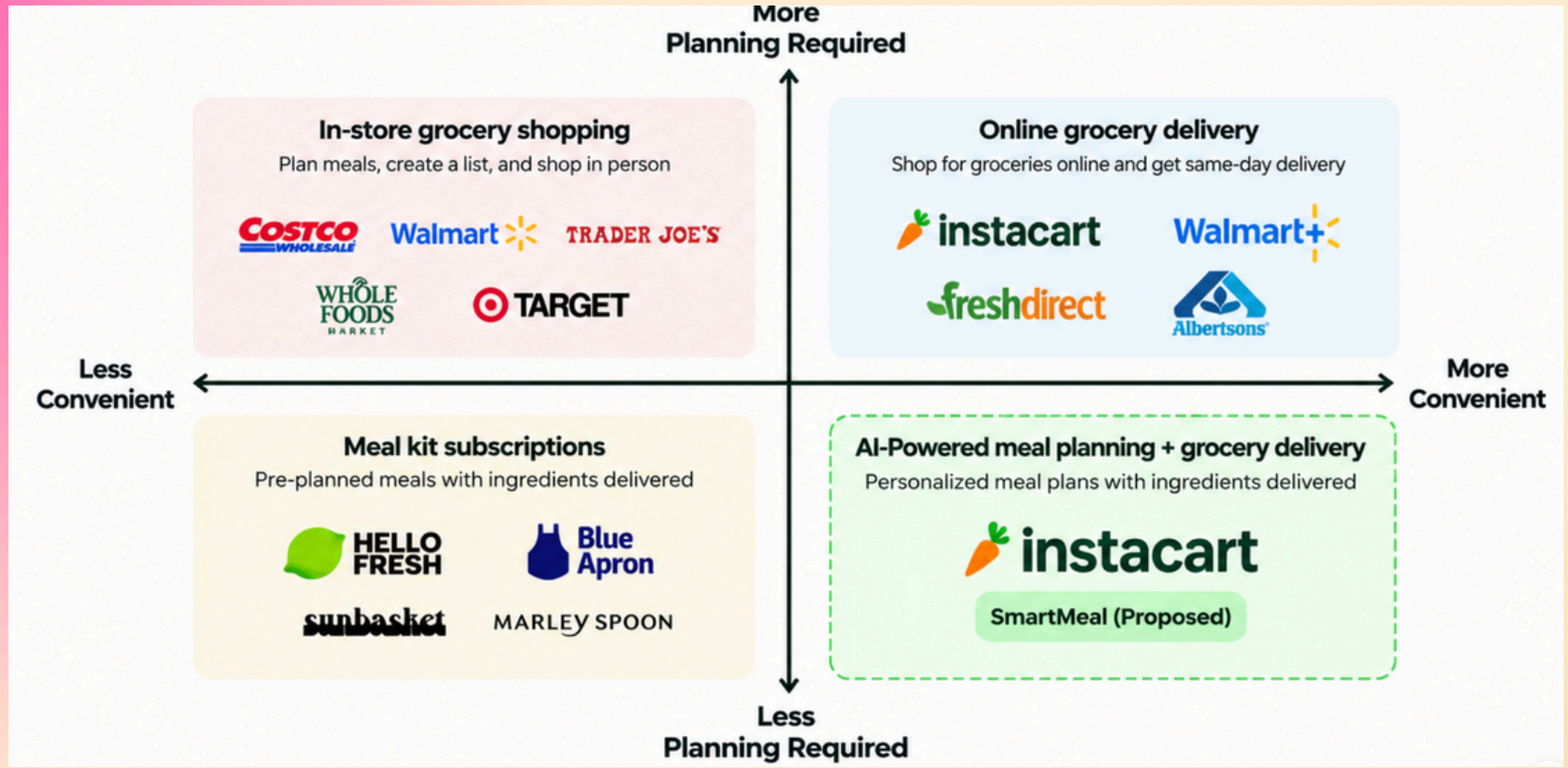
From Grocery Shopping to Smarter MealPlanning

How do people shop for groceries?

In today's fast paced world, People shop for groceries in different ways depending on their needs, budgets, and convenience. Instacart simplifies grocery shopping through three main services:

- Same day grocery delivery
- Online ordering with in store pickup
- Shopping from multiple retailers

In this case study, I explored whether Instacart should introduce a new experience: AI-Powered Meal Planning, helping users plan meals and automatically turn ingredients into grocery carts.

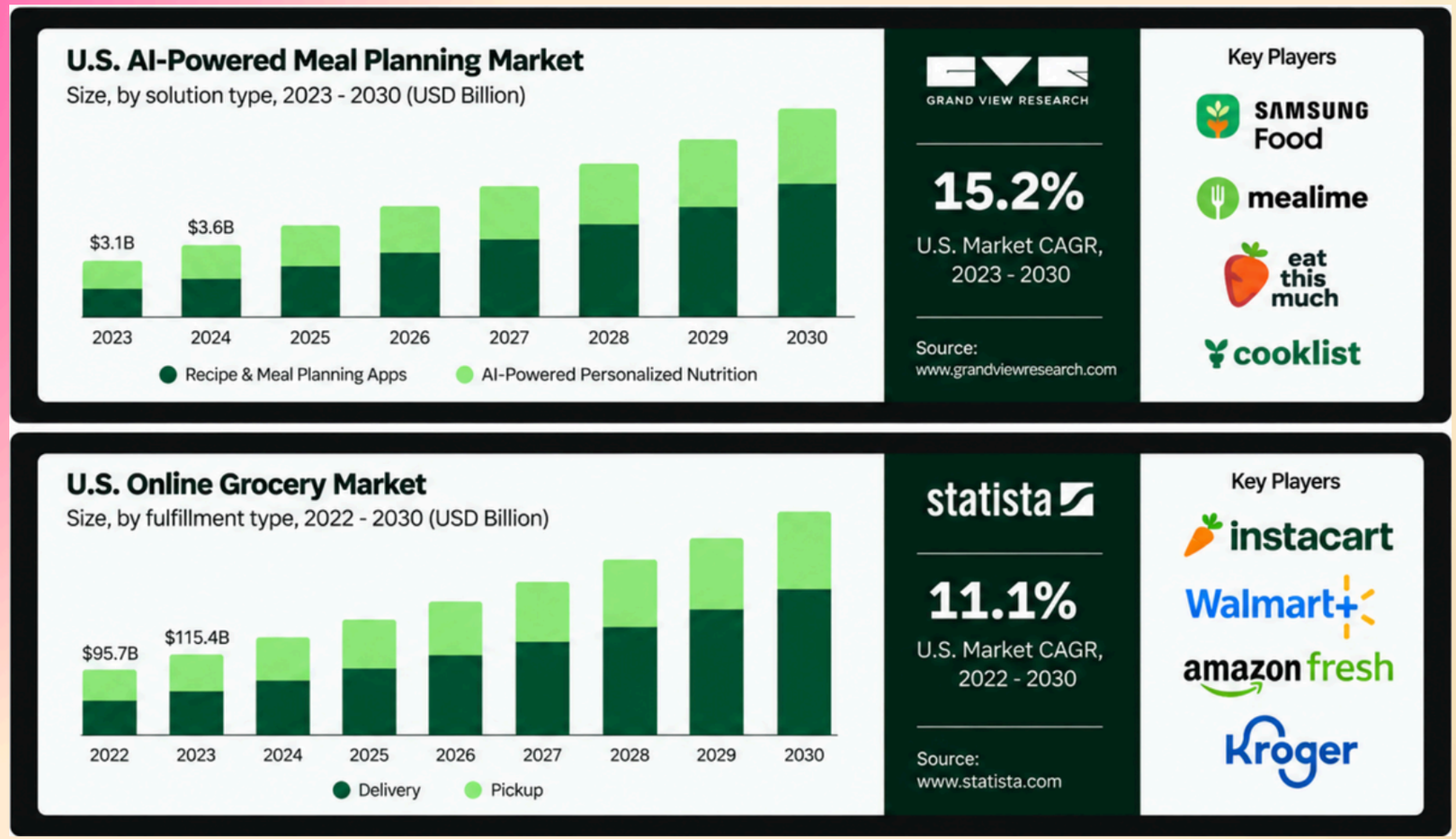


Hypothesis

If Instacart introduces AI-powered meal planning, then it will increase customer retention by simplifying weekly grocery shopping and encouraging repeat purchases.

Where is the AI-powered meal planning market headed?

The growing adoption of AI-powered personalization and digital grocery shopping creates an opportunity for smarter meal-planning solutions. As consumers seek convenience, affordability, and personalized experiences, AI-driven meal planning could become an increasingly valuable part of online grocery shopping.



Who are the users driving this growth?

Instacart shoppers and meal-planning app users share similar needs: convenience, affordability, and less time spent planning meals.

Instacart Users

Attitudes: Want convenient grocery shopping, flexible delivery, and access to their preferred stores.

Personas:

- Busy professionals (singles and couples)
- Working parents and families
- College students
- Older adults
- Budget-conscious households

AI Meal Planning Users*

Attitudes: Want personalized meal recommendations, healthier eating habits, budget control, and simplified grocery planning.

Personas:

- Busy professionals
- Health-conscious individuals
- Working parents
- College students
- People with dietary restrictions

These are hypothesized customer segments. Demographic overlap and customer attitudes require validation through research.

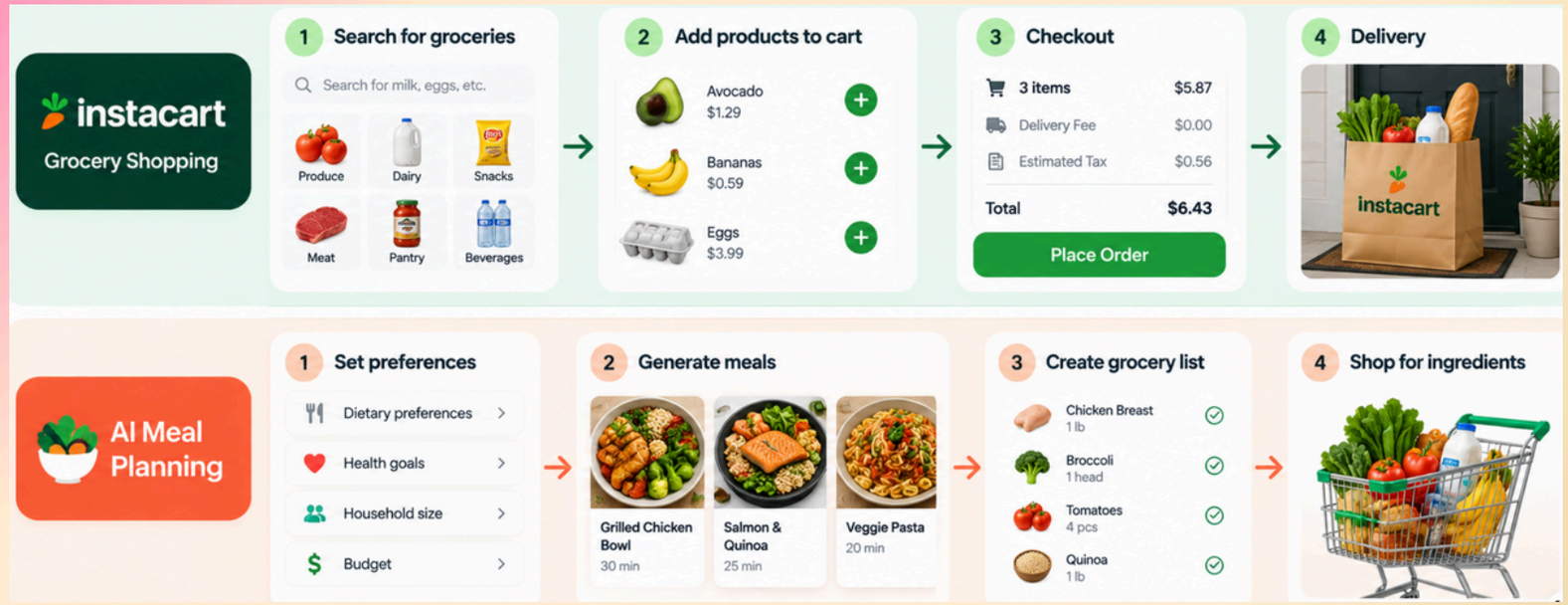
What are the pain points that these users need addressed?

1. Users want to save time: Planning meals, searching for recipes, and manually creating grocery lists can be time-consuming. AI-powered meal planning could simplify these tasks by generating personalized meal plans and grocery carts.

2. Users seek affordable grocery options: Customers want to stay within their budgets while purchasing nutritious meals. Smart meal planning could help estimate weekly grocery costs and recommend affordable ingredients.

3. Users want healthier meal choices: Customers may struggle to consistently plan balanced meals. Personalized recommendations could make it easier to find recipes aligned with dietary preferences and nutritional goals.

4. Users want to reduce food waste: Buying ingredients without a clear meal plan can lead to unused groceries. AI meal planning could recommend recipes that reuse ingredients across multiple meals.



How could AI-Powered Meal Planning affect Instacart's marketplace?

Since Instacart operates as a marketplace connecting customers, retailers, and shoppers, we should consider how introducing AI-powered meal planning could impact each group.

1. Grocery Retailers
Advantages

- Increased repeat grocery purchases
- Higher average basket sizes
- Improved product discovery through personalized recommendations
- Opportunities to promote complementary ingredients

Risks / Disadvantages

- Technical challenges with inventory and pricing integration
- Increased demand for certain ingredients, potentially causing stockouts
- AI recommendations may favor some products over others
- More substitutions if recommended ingredients are unavailable

2. Instacart Shoppers

Advantages

- Potential increase in weekly grocery orders
- Larger, more predictable shopping baskets
- Opportunities for more consistent order volume

Risks / Disadvantages

- Larger orders may require additional shopping time
- Unavailable ingredients could increase substitutions
- Complex ingredient lists may create fulfillment challenges
- Consolidated weekly orders could reduce the number of smaller orders

3. Grocery Brands

Advantages

- Increased product visibility through recipe recommendations
- Opportunities to introduce customers to new products
- Potential increase in recurring purchases
- Better discovery of complementary products

Risks / Disadvantages

- Reduced visibility if AI recommendations prioritize competing brands
- Increased competition for recommendation placement
- Potential shifts toward lower-cost alternatives

TL;DR: Instacart customers and marketplace partners could benefit from AI-Powered Meal Planning.

Instacart customers and meal-planning users share similar needs: convenience, affordability, personalization, and time savings.

- Customers could simplify weekly meal planning and grocery shopping.
- Retailers could benefit from recurring purchases and larger baskets.
- Grocery brands could gain visibility through personalized recommendations.
- Instacart could strengthen customer engagement and retention.

However, successful implementation would require accurate product matching, reliable inventory information, and a seamless shopping experience.

The Goal

Make Instacart SmartMeal part of customers' weekly routines, measured by 8-week retention of SmartMeal users compared with a control group.

MVP

To validate demand quickly, the pilot focuses on one core flow: plan → cart → checkout.

1. Set Preferences

As an Instacart user, I want to enter my household size, weekly budget, dietary preferences, and allergies so that my meal plan fits my life.
Allergies are treated as hard constraints: the AI can never recommend a meal containing a flagged allergen.

2. Generate a Weekly Meal Plan

As an Instacart user, I want a 5-meal weekly plan generated in one tap so that I spend less time deciding what to cook.
Users can swap any meal they don't like.

3. Pantry Check

As an Instacart user, I want to uncheck ingredients I already have so that I don't buy what's already in my kitchen.
This is the hardest problem in the flow. A recipe calls for 2 tbsp of soy sauce, but the store sells a full bottle. A quick pantry check prevents duplicate purchases and builds trust.

4. Auto-Build Cart

As an Instacart user, I want my remaining ingredients added to my cart from my preferred store so that I can check out in one step.
It uses Instacart's existing catalog, cart, and checkout without new infrastructure.

What options do we have to offer AI-Powered Meal Planning on Instacart?

1. Scheduled Meal Planning

Users can plan their weekly meals in advance and schedule grocery deliveries.
Order Types: Weekly Subscription or One-Time Order

Fulfillment Options:
Manual: Users generate meal plans and manually select ingredients from Instacart's grocery catalog.
Automated: AI converts meal plans into grocery carts, matches available products, and allows users to schedule deliveries through Instacart.

2. On-Demand Meal Planning

Users can instantly generate meal recommendations and order ingredients for same-day delivery.
Order Types: Single Meal, Multiple Meals, or Add-On

Fulfillment Options:
Instacart: AI recommends meals based on dietary preferences, budget, and available inventory. Ingredients are automatically added to the cart, allowing customers to review and place orders through Instacart's existing delivery network.

Which planning and fulfillment option works best?

Scheduled Weekly Meal Planning with Automated Grocery Cart Creation offers the strongest potential for customer convenience, recurring purchases, and long-term retention.

This would be my recommended option to test in an initial pilot.

	Scheduled		On-Demand
	Manual Meal Plan (User builds cart)	**Automated Meal Plan** (AI builds cart)	**Automated Meal Plan** (AI builds cart, same-day)
Business Impact	Medium	High	Medium
User Experience	Medium	High	High
Ease of Implementation	High	Medium	Low
Customer Retention Potential	Medium	High	Medium
Grocery Cart Accuracy	High	Medium	Medium
Retailer Integration	High	Medium	Medium
Operational Complexity	Low	Medium	High
Scalability	Medium	High	High
Budget Personalization	Medium	High	High
Repeat Purchase Potential	Medium	High	Medium

High / Favorable Medium Low / Challenging

Final Solution: Instacart SmartMeal

AI-Powered Weekly Meal Planning with Automated Grocery Cart Creation
This solution is:

- **Time-Saving: Generate personalized meal plans and grocery carts in minutes.**
- **Budget-Friendly: Plan weekly meals based on your grocery budget.**
- **Healthy Options: Discover meals tailored to your dietary preferences.**

- **Convenient: Schedule grocery deliveries directly through Instacart.**
- **Less Food Waste: Reuse ingredients across meals to minimize waste.**

The User Experience

- Separate Meal Planning Section: Introduce SmartMeal as a dedicated feature without disrupting Instacart's existing shopping experience.
- Simple and Intuitive: Guide users through meal selection, customization, and grocery checkout.
- Personalized Recommendations: Suggest meals based on dietary preferences, household size, and budget.
- Seamless Integration: Automatically convert meal plans into grocery carts using Instacart's existing shopping and checkout experience.
- Message the Benefits: Highlight time savings, personalized meals, and budget-friendly grocery shopping throughout the user journey.
- Use Existing Components: Leverage Instacart's existing product catalog, shopping cart, checkout, and delivery features for a familiar experience.

Phased Rollout of Instacart SmartMeal

- We will begin with a small-scale pilot to validate customer demand and product usability. Based on user feedback and performance metrics, we'll refine the experience before expanding.
- Who is involved? Launching SmartMeal would require collaboration across Product, Engineering, Design, Data Science, Operations, Marketing, Customer Support, and Retail Partnerships.
- Education: We'll guide customers through the new meal-planning experience and train internal teams to support the feature.

Pilot Metrics

NORTH STAR METRIC

8-Week Retention of SmartMeal Users vs. Control
The % of users still ordering groceries 8 weeks after first using SmartMeal, compared with similar users who didn't.

Leading Indicators

- Plan-to-cart conversion: % of generated plans that become carts. Does the plan feel useful?
- Cart-to-checkout conversion: % of SmartMeal carts that are purchased. Do users trust the AI-built cart?
- Week-2 repeat rate: % of users who generate a second plan. Is it a habit or a one-time novelty?
- Meal swap rate: % of meals replaced. A high rate means recommendations miss.

Lagging Indicators

- Average Revenue Per User (ARPU)
- 3-Month Customer Retention
- Repeat Purchase Rate

Guardrail Metrics

- Allergen violation rate: target 0. Any violation pauses the pilot.
- Item removal rate in cart: high removal means poor ingredient matching
- Substitution and refund rate: does SmartMeal make shopper fulfillment harder?
- Cart abandonment rate

One potential risk is that customers may consolidate multiple smaller orders into a single weekly purchase, reducing overall order frequency. We should monitor these metrics to ensure SmartMeal creates meaningful business value.

Monitoring Metrics

- AI Meal Generation Error Rate
- Ingredient Matching Accuracy
- Grocery Cart Creation Failure Rate
- Unavailable Product Rate

These metrics will help identify technical issues and ensure customers receive accurate meal recommendations and grocery carts.

Final Thoughts

In this case study, I explored whether introducing AI-Powered Meal Planning could increase Instacart's customer retention and repeat purchases.

My hypothesis was that simplifying meal planning and automatically generating grocery carts would reduce shopping friction while providing customers with a more personalized experience.

Instacart is well-positioned to explore this opportunity by leveraging its existing retailer partnerships, product catalog, and grocery delivery infrastructure.

As a Product Manager, I would recommend launching a small-scale pilot to evaluate customer adoption, meal plan-to-cart conversion, and repeat purchase behavior before expanding the feature.

The goal is simple: Make Instacart not just a place to buy groceries, but a smarter way to plan meals and shop.

What I Learned About Myself...

Working through this case study reminded me that the best product ideas don't always come from creating something entirely new, but from making everyday experiences simpler.

I also learned that building a valuable AI product requires more than an exciting feature. It means understanding customer problems, evaluating trade-offs, and measuring real business impact.

And as someone who occasionally spends more time deciding what to eat than actually cooking, I can definitely relate to this problem. 😂

If you made it all the way to the end, thank you so much for reading!